

Complex Variables — Solved Exercises

(Solved by the methods used in the notes)

1. Modulus and Argument

(i) $z = 2 + i$. $|z| = \sqrt{2^2 + 1^2} = \sqrt{5}$, $\arg z = \tan^{-1} \frac{1}{2} = 26.57^\circ$.

(ii) $z = 1 - 3i$. $|z| = \sqrt{1 + 9} = \sqrt{10}$, $\arg z = \tan^{-1} \left(\frac{-3}{1} \right) = -71.57^\circ$ (fourth quadrant).

(iii) $z = (1 + i)^2 = 1 + 2i + i^2 = 2i$, i.e. $x = 0$, $y = 2$. $|z| = \sqrt{0 + 4} = 2$. Since the point $(0, 2)$ lies on the positive imaginary axis, $\arg z = \frac{\pi}{2} = 90^\circ$.

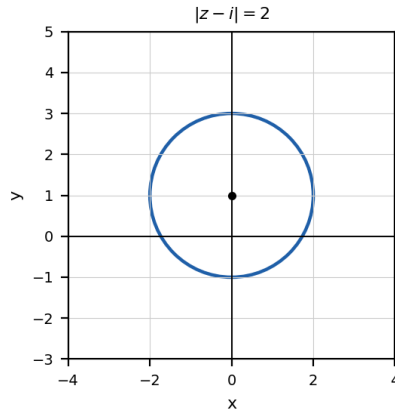
(iv) $z = \frac{2-i}{3+i}(1+i) = \frac{(2-i)(1+i)}{3+i} = \frac{2+2i-i-i^2}{3+i} = \frac{3+i}{3+i} = 1$. So $|z| = 1$ and $\arg z = \tan^{-1}(0) = 0$.

(v) $z = \frac{3+i}{1+i} = \frac{(3+i)(1-i)}{(1+i)(1-i)} = \frac{3-3i+i-i^2}{1-i^2} = \frac{4-2i}{2} = 2-i$. $|z| = \sqrt{4+1} = \sqrt{5}$, $\arg z = \tan^{-1} \left(\frac{-1}{2} \right) = -26.57^\circ$.

2. Geometrical description of regions

(i) $|z - i| = 2$. Put $z = x + iy$: $|x + iy - i| = 2 \Rightarrow |x + i(y-1)| = 2 \Rightarrow \sqrt{x^2 + (y-1)^2} = 2$. Squaring, $x^2 + (y-1)^2 = 2^2$.

This is a circle with centre $(0, 1)$ and radius 2.



(ii) $|z + 2i| + |z - 2i| = 10$.

$$\begin{aligned} \Rightarrow \sqrt{x^2 + (y+2)^2} + \sqrt{x^2 + (y-2)^2} &= 10 \\ \Rightarrow \sqrt{x^2 + (y+2)^2} &= 10 - \sqrt{x^2 + (y-2)^2} \end{aligned}$$

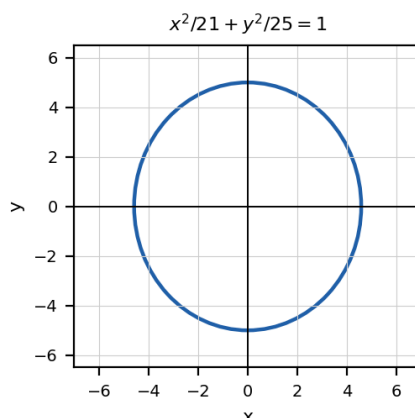
Squaring both sides, we get

$$\begin{aligned} x^2 + y^2 + 4y + 4 &= 100 - 20\sqrt{x^2 + (y-2)^2} + x^2 + y^2 - 4y + 4 \\ \Rightarrow 20\sqrt{x^2 + (y-2)^2} &= 100 - 8y \Rightarrow 5\sqrt{x^2 + (y-2)^2} = 25 - 2y \end{aligned}$$

Again squaring both sides, we get,

$$\begin{aligned}
 25\{x^2 + (y - 2)^2\} &= (25 - 2y)^2 \\
 \Rightarrow 25x^2 + 25y^2 - 100y + 100 &= 625 - 100y + 4y^2 \\
 \Rightarrow 25x^2 + 21y^2 &= 525 \Rightarrow \frac{x^2}{21} + \frac{y^2}{25} = 1 \Rightarrow \frac{x^2}{(\sqrt{21})^2} + \frac{y^2}{5^2} = 1
 \end{aligned}$$

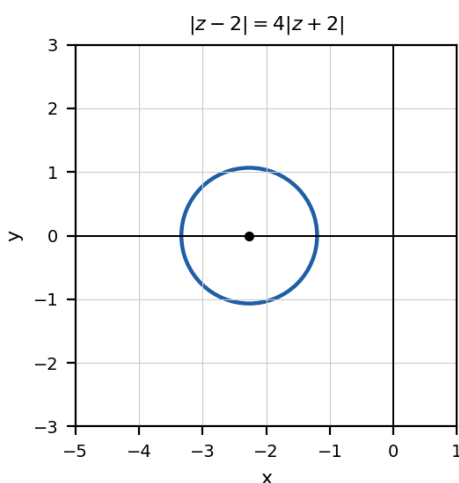
This represents an ellipse passing through the points $(\pm\sqrt{21}, 0)$ and $(0, \pm 5)$ (foci at $(0, \pm 2)$).



(iii) $|z - 2| = 4|z + 2|$. Put $z = x + iy$: $\sqrt{(x - 2)^2 + y^2} = 4\sqrt{(x + 2)^2 + y^2}$. Squaring both sides, we get,

$$\begin{aligned}
 (x - 2)^2 + y^2 &= 16\{(x + 2)^2 + y^2\} \\
 \Rightarrow x^2 - 4x + 4 + y^2 &= 16x^2 + 64x + 64 + 16y^2 \\
 \Rightarrow 15x^2 + 68x + 15y^2 + 60 &= 0 \Rightarrow x^2 + \frac{68}{15}x + y^2 + 4 = 0 \\
 \Rightarrow x^2 + 2 \cdot x \cdot \frac{34}{15} + \left(\frac{34}{15}\right)^2 + y^2 &= \left(\frac{34}{15}\right)^2 - 4 = \frac{1156 - 900}{225} = \frac{256}{225} \\
 \Rightarrow \left(x + \frac{34}{15}\right)^2 + y^2 &= \left(\frac{16}{15}\right)^2
 \end{aligned}$$

This represents a circle of radius $\frac{16}{15}$ and centre $\left(-\frac{34}{15}, 0\right)$.



(iv) $|z+2| + |z-2| = 3$.

$$\Rightarrow \sqrt{(x+2)^2 + y^2} + \sqrt{(x-2)^2 + y^2} = 3 \Rightarrow \sqrt{(x+2)^2 + y^2} = 3 - \sqrt{(x-2)^2 + y^2}$$

Squaring both sides, we get

$$\begin{aligned} x^2 + 4x + 4 + y^2 &= 9 - 6\sqrt{(x-2)^2 + y^2} + x^2 - 4x + 4 + y^2 \\ \Rightarrow 6\sqrt{(x-2)^2 + y^2} &= 9 - 8x \end{aligned}$$

Again squaring both sides, $36\{(x-2)^2 + y^2\} = (9-8x)^2 \Rightarrow 36x^2 - 144x + 144 + 36y^2 = 81 - 144x + 64x^2 \Rightarrow 28x^2 - 36y^2 = 63$, i.e. $\frac{x^2}{9/4} - \frac{y^2}{7/4} = 1$.

This equation does not describe the given locus: it forces $|x| \geq \frac{3}{2}$, whereas $6\sqrt{(x-2)^2 + y^2} = 9 - 8x \geq 0$ needs $x \leq \frac{9}{8}$, so $x \leq -\frac{3}{2}$; but then $\sqrt{(x-2)^2 + y^2} = \frac{9-8x}{6} \geq \frac{21}{6} > 3$, which contradicts $\sqrt{(x-2)^2 + y^2} = 3 - \sqrt{(x+2)^2 + y^2} \leq 3$. (The extra curve appears only because of squaring.) The same conclusion follows at once from the triangle inequality: $|z+2| + |z-2| \geq |(z+2) - (z-2)| = 4 > 3$.

Hence no point z satisfies the equation; the region is the empty set. (For an ellipse with foci ± 2 the constant must be greater than 4.)

(v) $\left| \frac{z-3}{z+3} \right| = 2 \Rightarrow |z-3| = 2|z+3|$. Put $z = x + iy$: $\sqrt{(x-3)^2 + y^2} = 2\sqrt{(x+3)^2 + y^2}$. Squaring both sides,

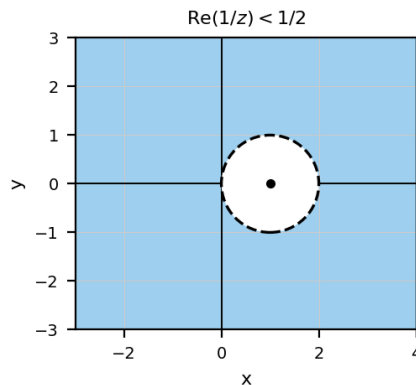
$$\begin{aligned} (x-3)^2 + y^2 &= 4\{(x+3)^2 + y^2\} \Rightarrow 3x^2 + 30x + 3y^2 + 27 = 0 \\ \Rightarrow x^2 + 10x + y^2 + 9 &= 0 \Rightarrow (x+5)^2 + y^2 = 16 = 4^2 \end{aligned}$$

This represents a circle of radius 4 and centre $(-5, 0)$ (same as the worked example in the notes).

(vi) $\operatorname{Re}\left(\frac{1}{z}\right) < \frac{1}{2}$. Let $z = x + iy$.

$$\begin{aligned} \Rightarrow \operatorname{Re}\left(\frac{1}{x+iy}\right) &< \frac{1}{2} \Rightarrow \operatorname{Re}\left\{\frac{x-iy}{(x+iy)(x-iy)}\right\} < \frac{1}{2} \Rightarrow \operatorname{Re}\left(\frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}\right) < \frac{1}{2} \\ \Rightarrow \frac{x}{x^2+y^2} &< \frac{1}{2} \Rightarrow 2x < x^2 + y^2 \Rightarrow x^2 + y^2 - 2x > 0 \Rightarrow (x-1)^2 + y^2 > 1 \end{aligned}$$

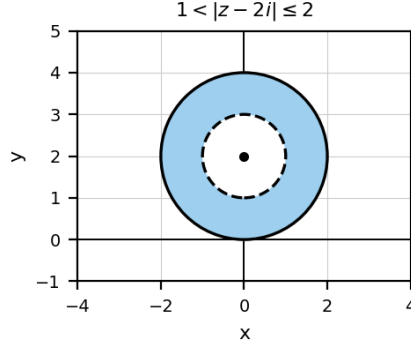
The region is the set of all external points of the circle with centre $(1, 0)$ and radius 1 (boundary not included).



(vii) $1 < |z-2i| \leq 2$. Let $z = x + iy$.

$$\begin{aligned} \Rightarrow 1 < |x + iy - 2i| &\leq 2 \Rightarrow 1 < |x + i(y-2)| \leq 2 \Rightarrow 1 < \sqrt{x^2 + (y-2)^2} \leq 2 \\ \Rightarrow 1^2 < x^2 + (y-2)^2 &\leq 2^2 \Rightarrow 1 < x^2 + (y-2)^2 \text{ and } x^2 + (y-2)^2 \leq 2^2 \end{aligned}$$

So $x^2 + (y - 2)^2 > 1$ represents all points external to the circle with centre $(0, 2)$ and radius 1, and $x^2 + (y - 2)^2 \leq 4$ represents all points internal and on the circle with centre $(0, 2)$ and radius 2. Thus the region is the set of all common points of the external points of the circle $x^2 + (y - 2)^2 = 1$ and the internal points of the circle $x^2 + (y - 2)^2 = 2^2$, including the boundary of the outer circle.



3. Cauchy's Integral Formula

1. Show that $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, $C : |z| = 5$.

Here $f(z) = \sin 3z$ is analytic inside and on $|z| = 5$. Also $z = a = -\frac{\pi}{2}$ and $|z| = \frac{\pi}{2} = 1.57 < 5$, so $z = -\frac{\pi}{2}$ lies inside C . By Cauchy's integral formula,

$$\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i f\left(-\frac{\pi}{2}\right) = 2\pi i \sin\left(-\frac{3\pi}{2}\right) = 2\pi i \cdot 1 = 2\pi i.$$

2. Evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$, $C : |z| = 4$.

$f(z) = e^{3z}$ is analytic inside and on C ; $z = a = -\pi i$, $|z| = \pi = 3.14 < 4$, so it lies inside C .

$$\oint_C \frac{e^{3z}}{z + \pi i} dz = 2\pi i f(-\pi i) = 2\pi i e^{-3\pi i} = 2\pi i (\cos 3\pi - i \sin 3\pi) = 2\pi i (-1 - 0) = -2\pi i.$$

3. Evaluate $\oint_C \frac{e^{-z}}{z - 1} dz$, $C : |z| = 2$.

$f(z) = e^{-z}$ is analytic inside and on C ; $z = a = 1$, $|1| = 1 < 2$, so it lies inside C .

$$\oint_C \frac{e^{-z}}{z - 1} dz = 2\pi i f(1) = 2\pi i e^{-1} = \frac{2\pi i}{e}.$$

4. Show that $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z - 1)(z - 2)} dz = 4\pi i$, $C : |z| = 3$.

This is the same integral as Question-2 of the worked examples. Using $\frac{1}{(z - 1)(z - 2)} = \frac{1}{z - 2} - \frac{1}{z - 1}$ and $f(z) = \sin \pi z^2 + \cos \pi z^2$ (analytic; $z = 1, 2$ inside C):

$$\oint_C \frac{f(z)}{z - 2} dz = 2\pi i f(2) = 2\pi i (\sin 4\pi + \cos 4\pi) = 2\pi i, \quad \oint_C \frac{f(z)}{z - 1} dz = 2\pi i f(1) = 2\pi i (\sin \pi + \cos \pi) = -2\pi i.$$

Hence the integral $= 2\pi i - (-2\pi i) = 4\pi i$.

4. Harmonic Functions and Conjugates

Exercise 1. $u = x^2 - y^2 - 2xy - 2x + 3y$.

$$\begin{aligned}\frac{\partial u}{\partial x} &= 2x - 2y - 2 = \varphi_1(x, y)(1) & \frac{\partial^2 u}{\partial x^2} &= 2(2) \\ \frac{\partial u}{\partial y} &= -2y - 2x + 3 = \varphi_2(x, y)(3) & \frac{\partial^2 u}{\partial y^2} &= -2(4)\end{aligned}$$

Adding (2) and (4): $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0$. So u satisfies the Laplace equation and is harmonic.

Putting $x = z$, $y = 0$ in (1) and (3): $\varphi_1(z, 0) = 2z - 2$, $\varphi_2(z, 0) = -2z + 3$. By Milne's method,

$$f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = 2z - 2 - i(-2z + 3) = (2 + 2i)z - (2 + 3i).$$

$\int f'(z)dz \Rightarrow f(z) = (1 + i)z^2 - (2 + 3i)z + c_1 + ic_2$. Then

$$\begin{aligned}u + iv &= (1 + i)(x + iy)^2 - (2 + 3i)(x + iy) + c_1 + ic_2 \\ &= (1 + i)(x^2 - y^2 + 2ixy) - (2x + 2iy + 3ix - 3y) + c_1 + ic_2 \\ &= (x^2 - y^2 - 2xy) + i(x^2 - y^2 + 2xy) - 2x + 3y - i(3x + 2y) + c_1 + ic_2 \\ &= (x^2 - y^2 - 2xy - 2x + 3y) + c_1 + i(x^2 - y^2 + 2xy - 3x - 2y + c_2)\end{aligned}$$

Equating imaginary parts, $\boxed{v = x^2 - y^2 + 2xy - 3x - 2y + c_2}$.

Exercise 2. $u = 2x - x^3 + 3xy^2$.

$$\begin{aligned}\frac{\partial u}{\partial x} &= 2 - 3x^2 + 3y^2 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= -6x(2) \\ \frac{\partial u}{\partial y} &= 6xy = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= 6x(4)\end{aligned}$$

(2)+(4): $-6x + 6x = 0$, so u is harmonic. Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 2 - 3z^2$, $\varphi_2(z, 0) = 0$.

$$f'(z) = 2 - 3z^2 \Rightarrow f(z) = 2z - z^3 + c_1 + ic_2.$$

$$\begin{aligned}u + iv &= 2(x + iy) - (x + iy)^3 + c_1 + ic_2 = 2x + 2iy - (x^3 + 3ix^2y - 3xy^2 - iy^3) + c_1 + ic_2 \\ &= (2x - x^3 + 3xy^2) + c_1 + i(2y - 3x^2y + y^3 + c_2)\end{aligned}$$

Hence $\boxed{v = 2y - 3x^2y + y^3 + c_2}$.

Exercise 3. $u = e^x \cos y$.

$$\begin{aligned}\frac{\partial u}{\partial x} &= e^x \cos y = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= e^x \cos y(2) \\ \frac{\partial u}{\partial y} &= -e^x \sin y = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= -e^x \cos y(4)\end{aligned}$$

(2)+(4): $e^x \cos y - e^x \cos y = 0$, so u is harmonic. Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = e^z$, $\varphi_2(z, 0) = 0$. Then $f'(z) = e^z \Rightarrow f(z) = e^z + c_1 + ic_2$.

$$u + iv = e^{x+iy} + c_1 + ic_2 = e^x(\cos y + i \sin y) + c_1 + ic_2 = e^x \cos y + c_1 + i(e^x \sin y + c_2).$$

Hence $\boxed{v = e^x \sin y + c_2}$.

Exercise 4. Yes. Let $u = c$ (a constant). Then $u_x = u_y = 0$, so $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 + 0 = 0$; the Laplace equation is satisfied, so a constant function is (trivially) harmonic. Its harmonic conjugate is also a constant (by the C-R equations $v_y = u_x = 0$, $v_x = -u_y = 0$).

Exercise 5. $u = x^2 + y^2$: $\frac{\partial^2 u}{\partial x^2} = 2$, $\frac{\partial^2 u}{\partial y^2} = 2$, so $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 4 \neq 0$. The Laplace equation is not satisfied, so $u = x^2 + y^2$ is **not** harmonic.

Exercise 6. $u = x^2 - y^2$: $\frac{\partial u}{\partial x} = 2x = \varphi_1$, $\frac{\partial^2 u}{\partial x^2} = 2$; $\frac{\partial u}{\partial y} = -2y = \varphi_2$, $\frac{\partial^2 u}{\partial y^2} = -2$. Sum $= 2 - 2 = 0$, so u is harmonic.

$\varphi_1(z, 0) = 2z$, $\varphi_2(z, 0) = 0$, so $f'(z) = 2z \Rightarrow f(z) = z^2 + c_1 + ic_2$. Then $u + iv = (x + iy)^2 + c_1 + ic_2 = x^2 - y^2 + c_1 + i(2xy + c_2)$. Hence the harmonic conjugate is $\boxed{v = 2xy + c_2}$.

Exercise 7. $u = x^2 - y^2 - 4x + 6y$.

$$\begin{array}{ll} \frac{\partial u}{\partial x} = 2x - 4 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} = 2(2) \\ \frac{\partial u}{\partial y} = -2y + 6 = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} = -2(4) \end{array}$$

(2)+(4): $2 - 2 = 0$, so u is harmonic. Putting $x = z, y = 0$: $\varphi_1(z, 0) = 2z - 4$, $\varphi_2(z, 0) = 6$.

$$f'(z) = 2z - 4 - 6i \Rightarrow f(z) = z^2 - (4 + 6i)z + c_1 + ic_2.$$

$$\begin{aligned} u + iv &= (x + iy)^2 - (4 + 6i)(x + iy) + c_1 + ic_2 = x^2 - y^2 + 2ixy - 4x - 4iy - 6ix + 6y + c_1 + ic_2 \\ &= (x^2 - y^2 - 4x + 6y) + c_1 + i(2xy - 6x - 4y + c_2) \end{aligned}$$

Hence $\boxed{v = 2xy - 6x - 4y + c_2}$.

Exercise 8. $u = 2x^2 - 4y^2 - 5xy - 4x + y$.

$$\frac{\partial u}{\partial x} = 4x - 5y - 4, \quad \frac{\partial^2 u}{\partial x^2} = 4; \quad \frac{\partial u}{\partial y} = -8y - 5x + 1, \quad \frac{\partial^2 u}{\partial y^2} = -8$$

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 4 - 8 = -4 \neq 0$. So u does not satisfy the Laplace equation: u is **not harmonic**, and therefore no harmonic conjugate $v(x, y)$ exists.

5. Residue Theorem Exercises

1. Show that $\oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{i}{\pi}$, $C : |z| = 4$.

The poles are given by $(z + \pi i)^2(z - \pi i)^2 = 0$, i.e. $z = \pm \pi i$, both double poles, and $|\pm \pi i| = \pi < 4$, so both lie inside C . Using the residues found in Question-3 of the notes (with $e^{\pi i} = e^{-\pi i} = -1$):

$$\text{Res}_{z=\pi i} = -\frac{(\pi + i)e^{\pi i}}{4\pi^3} = \frac{\pi + i}{4\pi^3}, \quad \text{Res}_{z=-\pi i} = -\frac{(\pi - i)e^{-\pi i}}{4\pi^3} = \frac{\pi - i}{4\pi^3}.$$

By Cauchy's residue theorem,

$$\oint_C f(z) dz = 2\pi i \left[\frac{\pi + i}{4\pi^3} + \frac{\pi - i}{4\pi^3} \right] = 2\pi i \cdot \frac{2\pi}{4\pi^3} = 2\pi i \cdot \frac{1}{2\pi^2} = \frac{i}{\pi}.$$

2. Evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$, $C : |z + 1| = 4$.

The pole is $z = -\pi i$ (simple). $|\pi i + 1| = \sqrt{1 + \pi^2} \approx 3.30 < 4$, so it lies inside C . Residue
 $= \lim_{z \rightarrow -\pi i} (z + \pi i) \frac{e^{3z}}{z + \pi i} = e^{-3\pi i} = \cos 3\pi - i \sin 3\pi = -1$. Therefore

$$\oint_C \frac{e^{3z}}{z + \pi i} dz = 2\pi i(-1) = -2\pi i.$$

3. Evaluate $\oint_C \frac{z}{(z^2 + 1)(z - 3)} dz$, $C : |z| = 2$.

Poles: $z = i, -i, 3$ (all simple). $|\pm i| = 1 < 2$ (inside), $|3| = 3 > 2$ (outside). So only $z = \pm i$ are used.

$$\text{Res}_{z=i} = \lim_{z \rightarrow i} (z - i) \frac{z}{(z + i)(z - i)(z - 3)} = \frac{i}{2i(i - 3)} = \frac{1}{2(i - 3)} = \frac{-3 - i}{20}$$

$$\text{Res}_{z=-i} = \lim_{z \rightarrow -i} (z + i) \frac{z}{(z + i)(z - i)(z - 3)} = \frac{-i}{(-2i)(-i - 3)} = \frac{1}{2(-i - 3)} = \frac{-3 + i}{20}$$

Sum of residues $= \frac{-6}{20} = -\frac{3}{10}$. By Cauchy's residue theorem,

$$\oint_C \frac{z}{(z^2 + 1)(z - 3)} dz = 2\pi i \left(-\frac{3}{10} \right) = -\frac{3\pi i}{5}.$$

4. Evaluate $\oint_C \frac{z + 1}{z^2 - 2z} dz$, $C : |z| = 3$.

$z^2 - 2z = z(z - 2)$, so the poles are $z = 0$ and $z = 2$ (simple), both inside $|z| = 3$.

$$\text{Res}_{z=0} = \lim_{z \rightarrow 0} z \cdot \frac{z + 1}{z(z - 2)} = \frac{1}{-2} = -\frac{1}{2}, \quad \text{Res}_{z=2} = \lim_{z \rightarrow 2} (z - 2) \frac{z + 1}{z(z - 2)} = \frac{3}{2}$$

Sum $= -\frac{1}{2} + \frac{3}{2} = 1$. Hence $\oint_C \frac{z + 1}{z^2 - 2z} dz = 2\pi i(1) = 2\pi i$.

6. Laurent Series Exercises

1. $f(z) = \frac{1}{z(z - 1)}$. Poles at $z = 0$ and $z = 1$, so the function is analytic in $0 < |z| < 1$ and in $|z| > 1$ (see the remark in the notes: $0 < |z| < 3$ contains the pole $z = 1$). Here

$$\frac{1}{z(z - 1)} = \frac{1}{z - 1} - \frac{1}{z}.$$

Region (i). The ring $0 < |z| < 3$ is split at $|z| = 1$ into the two valid rings:

(a) $0 < |z| < 1$: $|z| < 1$, so

$$\begin{aligned} f(z) &= -\frac{1}{1 - z} - \frac{1}{z} = -(1 - z)^{-1} - \frac{1}{z} = -(1 + z + z^2 + z^3 + \cdots) - \frac{1}{z} \\ &= -\frac{1}{z} - 1 - z - z^2 - z^3 - \cdots \end{aligned}$$

(b) $1 < |z| < 3$: $\frac{1}{|z|} < 1$, so

$$\begin{aligned} f(z) &= \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}} - \frac{1}{z} = \frac{1}{z} \left(1 - \frac{1}{z} \right)^{-1} - \frac{1}{z} \\ &= \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) - \frac{1}{z} = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \cdots \end{aligned}$$

Region (ii) $|z| > 2$ (which lies in $|z| > 1$, so $\frac{1}{|z|} < 1$): the same expansion applies,

$$f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots$$

2. $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$. Let $\frac{z^2 - 1}{(z+2)(z+3)} = 1 + \frac{A}{z+2} + \frac{B}{z+3}$, so

$$z^2 - 1 = (z+2)(z+3) + A(z+3) + B(z+2).$$

Putting $z = -2$: $3 = A$. Putting $z = -3$: $8 = -B \Rightarrow B = -8$. Therefore

$$f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}.$$

(i) $2 < |z| < 3$: $\left|\frac{2}{z}\right| < 1$ and $\left|\frac{z}{3}\right| < 1$.

$$\begin{aligned} f(z) &= 1 + \frac{3}{z} \cdot \frac{1}{1 + \frac{2}{z}} - \frac{8}{3} \cdot \frac{1}{1 + \frac{z}{3}} = 1 + \frac{3}{z} \left(1 + \frac{2}{z}\right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1} \\ &= 1 + \frac{3}{z} \left(1 - \frac{2}{z} + \frac{2^2}{z^2} - \frac{2^3}{z^3} + \dots\right) - \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{3^2} - \frac{z^3}{3^3} + \dots\right) \\ &= 1 + \left(\frac{3}{z} - \frac{6}{z^2} + \frac{12}{z^3} - \frac{24}{z^4} + \dots\right) - \left(\frac{8}{3} - \frac{8z}{9} + \frac{8z^2}{27} - \frac{8z^3}{81} + \dots\right) \\ &= \dots - \frac{24}{z^4} + \frac{12}{z^3} - \frac{6}{z^2} + \frac{3}{z} - \frac{5}{3} + \frac{8z}{9} - \frac{8z^2}{27} + \frac{8z^3}{81} - \dots \end{aligned}$$

(ii) $|z| < 2$: $\left|\frac{z}{2}\right| < 1$ and $\left|\frac{z}{3}\right| < 1$.

$$\begin{aligned} f(z) &= 1 + \frac{3}{2} \cdot \frac{1}{1 + \frac{z}{2}} - \frac{8}{3} \cdot \frac{1}{1 + \frac{z}{3}} = 1 + \frac{3}{2} \left(1 + \frac{z}{2}\right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1} \\ &= 1 + \frac{3}{2} \left(1 - \frac{z}{2} + \frac{z^2}{2^2} - \frac{z^3}{2^3} + \dots\right) - \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{3^2} - \frac{z^3}{3^3} + \dots\right) \\ &= \left(1 + \frac{3}{2} - \frac{8}{3}\right) + \left(-\frac{3}{4} + \frac{8}{9}\right)z + \left(\frac{3}{8} - \frac{8}{27}\right)z^2 + \left(-\frac{3}{16} + \frac{8}{81}\right)z^3 + \dots \\ &= -\frac{1}{6} + \frac{5}{36}z + \frac{17}{216}z^2 - \frac{115}{1296}z^3 + \dots \end{aligned}$$

3. $f(z) = \frac{1}{(z^2+1)(z+2)}$. Let $\frac{1}{(z^2+1)(z+2)} = \frac{Az+B}{z^2+1} + \frac{C}{z+2}$, so $1 = (Az+B)(z+2) + C(z^2+1)$. Putting $z = -2$: $1 = 5C \Rightarrow C = \frac{1}{5}$. Comparing the coefficients of z^2 : $A + C = 0 \Rightarrow A = -\frac{1}{5}$; constant terms: $2B + C = 1 \Rightarrow B = \frac{2}{5}$. Hence

$$f(z) = \frac{2-z}{5(z^2+1)} + \frac{1}{5(z+2)}.$$

(i) $1 < |z| < 2$: $\left|\frac{1}{z^2}\right| < 1$ and $\left|\frac{z}{2}\right| < 1$.

$$\begin{aligned} f(z) &= \frac{2-z}{5} \cdot \frac{1}{z^2} \left(1 + \frac{1}{z^2}\right)^{-1} + \frac{1}{10} \left(1 + \frac{z}{2}\right)^{-1} \\ &= \frac{2-z}{5} \left(\frac{1}{z^2} - \frac{1}{z^4} + \frac{1}{z^6} - \dots\right) + \frac{1}{10} \left(1 - \frac{z}{2} + \frac{z^2}{2^2} - \frac{z^3}{2^3} + \dots\right) \\ &= \frac{1}{5} \left(-\frac{1}{z} + \frac{2}{z^2} + \frac{1}{z^3} - \frac{2}{z^4} - \frac{1}{z^5} + \frac{2}{z^6} + \dots\right) + \frac{1}{10} - \frac{z}{20} + \frac{z^2}{40} - \frac{z^3}{80} + \dots \end{aligned}$$

(ii) $|z| > 2$ (the pole $z = -2$ lies in $|z| > 1$, so the valid ring is $|z| > 2$; here $\left|\frac{1}{z^2}\right| < 1$ and $\left|\frac{2}{z}\right| < 1$).

Now $\frac{1}{5(z+2)} = \frac{1}{5z} \left(1 + \frac{2}{z}\right)^{-1}$, so

$$\begin{aligned} f(z) &= \frac{1}{5} \left(-\frac{1}{z} + \frac{2}{z^2} + \frac{1}{z^3} - \frac{2}{z^4} - \frac{1}{z^5} + \frac{2}{z^6} + \cdots \right) + \frac{1}{5z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \frac{16}{z^4} - \frac{32}{z^5} + \cdots \right) \\ &= \frac{1}{5} \left(\frac{5}{z^3} - \frac{10}{z^4} + \frac{15}{z^5} - \frac{30}{z^6} + \cdots \right) = \frac{1}{z^3} - \frac{2}{z^4} + \frac{3}{z^5} - \frac{6}{z^6} + \cdots \end{aligned}$$